

Qatar-4b

R-2673

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_181015 · created 2026-08-30T01:22:09+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458406.86092 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.028
Transit depth (Rp/Rs)^2	3.26 +/- 0.99 [%]
Orbital Inclination (inc)	87.11 +/- 1.53
Airmass coefficient 1 (a1)	1.101 +/- 0.023
Airmass coefficient 2 (a2)	-0.063 +/- 0.023
Scatter in the residuals of the lightcurve fit is	2.13 %
Best Comparison Star	#3 - [376, 161]
Optimal Method	PSF photometry
Transit Duration (day)	0.0566 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.181 ± 0.028 vs 0.138 (deviation 1.54σ)
Duration fit vs published	0.0566 d vs 0.0892 d (deviation 11.23σ)
Mid-time O-C	1.0 min
Depth SNR	6.5
Residual scatter	2.13 %

Light curve

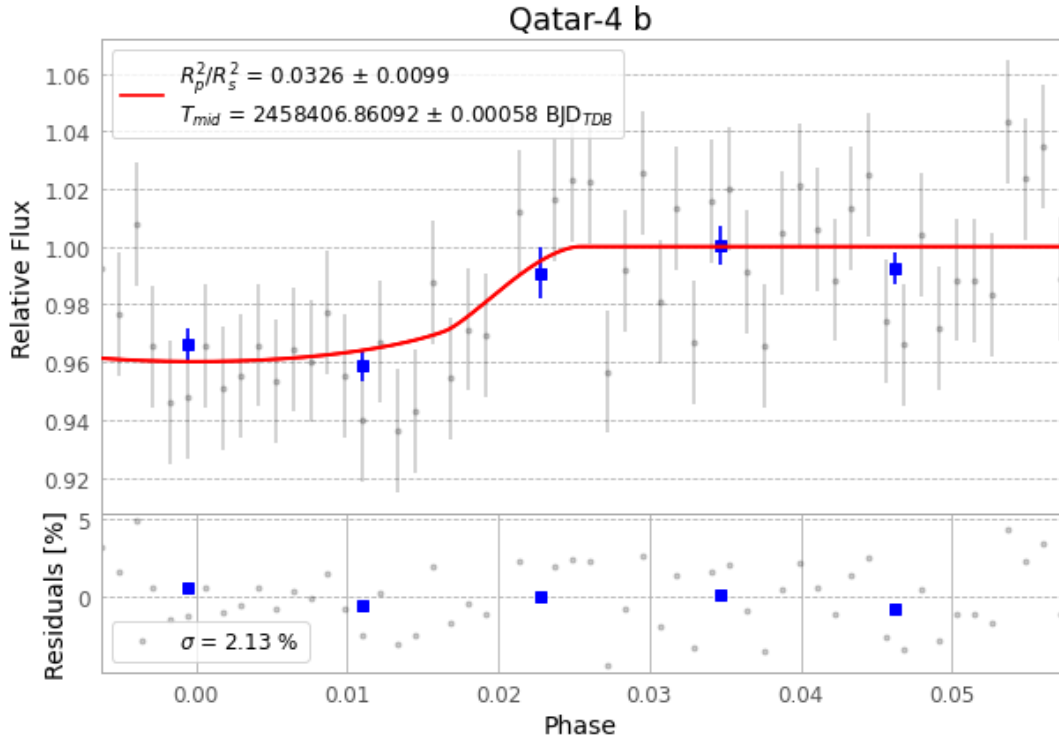


Figure 1 — FinalLightCurve_Qatar-4 b_2018-10-15.png (SHA-256 f7e4f60d30a8c4b7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [263, 373], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 367b352c7fe261bc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), Qatar-4181015081512.FITS → Qatar-4181015110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-181015.log (b33bec8c32a0...), FinalLightCurve_Qatar-4 b_2018-10-15.png (f7e4f60d30a8...), FinalLightCurve_Qatar-4 b_2018-10-15.csv (15ffe12ed26b...)

Compute resources

Wall time 92.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 01:22Z.